

BRAINDOUGH: A Single-Run TRIBE v2 Predicted-Response Artifact with Vertex-Index Brain Schematic Visualization

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Abstract

TRIBE v2 is a multimodal brain-encoding model that predicts average-subject fMRI-like cortical responses to video, audio, and language stimuli. This technical preprint reports a deliberately bounded BRAINDOUGH run that converts generated visual stimuli into short video events, executes one real TRIBE v2 prediction, checks the resulting artifact, and adds a transparent vertex-index brain schematic for review. The run generated 39 stimuli as context across five suites and retained one model response for `image_activation:color_gradient:static`. The response array has shape $2 \times 20,484$, corresponding to two retained model output segments over an fsaverage5-like cortical vertex vector; mean absolute predicted response was 0.102670. Because the artifact does not include fsaverage5 surface coordinates or an ROI atlas, the visualization uses an explicit convention: split the vector into two 10,242-vertex halves and project values onto a two-hemisphere schematic. The central contribution is a reproducible local artifact and reporting pattern, not a measured neuroscience result or spatial-registration method: all values are model-predicted average-subject responses, the brain figure is a vertex-index schematic rather than anatomical registration, and no subject-specific, diagnostic, hemispheric, or ROI claim is made.

1 Introduction

Foundation-model approaches are beginning to make in-silico neuroscience workflows practical on local workstations. TRIBE v2 is a recent tri-modal brain encoding model for video, audio, and language that predicts fMRI-like responses to naturalistic and experimental stimuli [1]. The public repository exposes pretrained-model loading, event extraction from video, and predictions shaped as `(n_timesteps, n_vertices)` [2]. The repository further states that predictions are for an “average” subject on an fsaverage5 cortical mesh and are shifted to compensate for hemodynamic lag [2]. The model weights are distributed on Hugging Face under a non-commercial Creative Commons license [3].

BRAINDOUGH is a local-first harness for small, reproducible brain-modeling experiments. Its practical design constraint is that source code, configs, and small derivative reports remain in the repository, while model weights, datasets, caches, and run outputs live under a separate `BRAINDOUGH_HOME`. This separation matters for TRIBE v2 because the checkpoint and feature extractors are large, local inference is slow enough to require explicit budgets, and casual visualization can easily overstate what the model output means.

This manuscript is a methods note for a single bounded run. It answers three engineering and reporting questions:

1. Can a clean BRAINDOUGH checkout execute a real, capped TRIBE v2 run and produce a structurally valid artifact?

2. Can the artifact be summarized in a reviewable HTML/PDF report without vendoring heavy response arrays or model weights into the repository?
3. Can a brain diagram be added without claiming anatomical ROI mapping when only a 20,484-value vertex vector is available?

2 Run Artifact

The source run is 20260516T230219131728Z-local-tribe-v2-first-suite-tribe-v2-eac66ffd. It was produced by the root command `just run-tribe` from the local first-suite spec `experiments/local/tribe_v2_first_suite.yaml`. The spec uses `backend: tribe-v2`, checkpoint `facebook/tribev2`, a 50-second event window, and `max_predictions: 1`. That cap is intentional: the first-suite generator creates more media than this laptop-scale local run should fully score.

The run completed on 2026-05-16 and passed structural artifact validation with `braindough validate`. It generated 39 stimuli across five suites. Only the first image-activation stimulus was passed through the real TRIBE v2 prediction budget; the other 38 generated stimuli are unscored run context. Table 1 gives the run-level facts copied from `metrics.json` and `response_metrics.csv`. Artifact validation means file, schema, and internal-consistency validation, not validation of TRIBE v2 calibration, accuracy, or neuroscientific interpretation.

Table 1: Run-level summary for the source run.

Field	Value
Backend	<code>tribe-v2</code>
Checkpoint	<code>facebook/tribev2</code>
Generated stimuli	39
Attempted predictions	1
Completed model predictions	1
Predicted stimulus	<code>image_activation:color_gradient:static</code>
Response shape	$2 \times 20,484$
Mean absolute predicted response	0.102670
Mean predicted response	-0.075532
Standard deviation	0.148002
Device ladder	<code>mps</code> , then <code>cpu</code>

3 Methods

3.1 Stimulus temporalization

TRIBE v2 inference is event-based. For static generated images, BRAINDOUGH creates short silent MP4 clips and submits the video path to the TRIBE v2 event extractor. This matches the public quick-start interface, which derives events from a `video_path` before prediction [2]. The local first suite generated baseline image-activation videos, visual controls, visual perturbations, temporalization variants, and audio controls. The real prediction budget kept only the first baseline image-activation clip.



Figure 1: Generated source image converted into a short silent video for the retained TRIBE v2 prediction. The video event, not the still image directly, is the model input.

The predicted clip was generated from a deterministic 256 by 256 RGB `color_gradient.png` image. BRAINDOUGH wrote an MP4 with eight identical RGB frames, 4 frames per second, 2 seconds duration, H.264 `libx264` encoding, macroblock size 16, and imageio quality 6. The TRIBE v2 backend then represented that clip as a 50-second video event before prediction, matching the local spec’s `event_duration_seconds: 50.0`.

3.2 Response summarization

The retained model-predicted response is stored externally as `outputs/responses.npz` with index metadata in `outputs/responses.index.jsonl`. The copied manuscript figures summarize that response without committing the response array itself. The primary scalar summary is mean absolute predicted response across the two retained model segments and all vertices. In this run, TRIBE v2 reported 100 possible prediction segments and the wrapper retained two segments under the configured 50-second event window and mask. This manuscript treats the two rows as model output segments and averages only for artifact summarization; it does not interpret them as independent measured time points. Additional summaries show the vertex-wise profile, the distribution of predicted values, and per-segment statistics.

3.3 Vertex-index brain schematic

TRIBE’s public documentation says predictions live on `fsaverage5` at roughly 20k vertices [2]. Surface-based coordinate systems such as FreeSurfer’s `fsaverage` family derive from cortical-surface registration methods designed to align folded cortical geometry across subjects [4, 5]. A true anatomical rendering would therefore require the matching surface coordinates or an atlas/annotation file. This run artifact has neither.

The added brain diagram consequently uses a conservative schematic label: *fsaverage5 vertex-index schematic*. The 20,484 response values are split into two 10,242-value halves and labeled

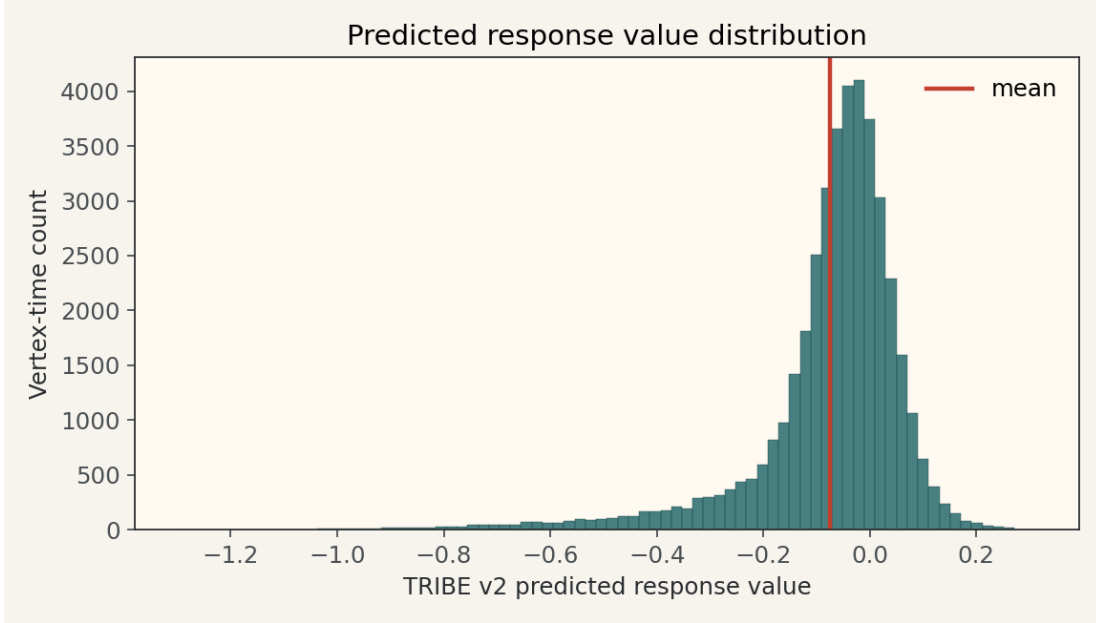


Figure 2: Distribution of predicted response values across both retained segments and all 20,484 vertices.

left/right only as an index convention. Without the matching TRIBE v2 output specification, surface coordinates, or annotation files, the split should not be interpreted as independently verified hemispheric anatomy. Each half is deterministically projected into a two-dimensional cortical silhouette. Color encodes mean absolute predicted response across the two retained model output segments. The figure highlights a small number of high-amplitude indices for visual QA, but it does not assign anatomical areas or ROIs.

Table 2: Brain-schematic metadata copied and normalized from the run-local schematic metadata file.

Field	Value
Schematic type	<code>fsaverage5_vertex_index_schematic</code>
First-half index range	0–10,241
Second-half index range	10,242–20,483
First-half mean absolute predicted response	0.102728
Second-half mean absolute predicted response	0.102612
First-half maximum absolute predicted response	1.001841
Second-half maximum absolute predicted response	1.078459
Color scale 2nd percentile	0.006217
Color scale 99.5th percentile	0.781990

4 Results

The result is intentionally small. The run demonstrates that the local harness can prepare generated video stimuli, load the cached TRIBE v2 checkpoint, extract video features, produce one

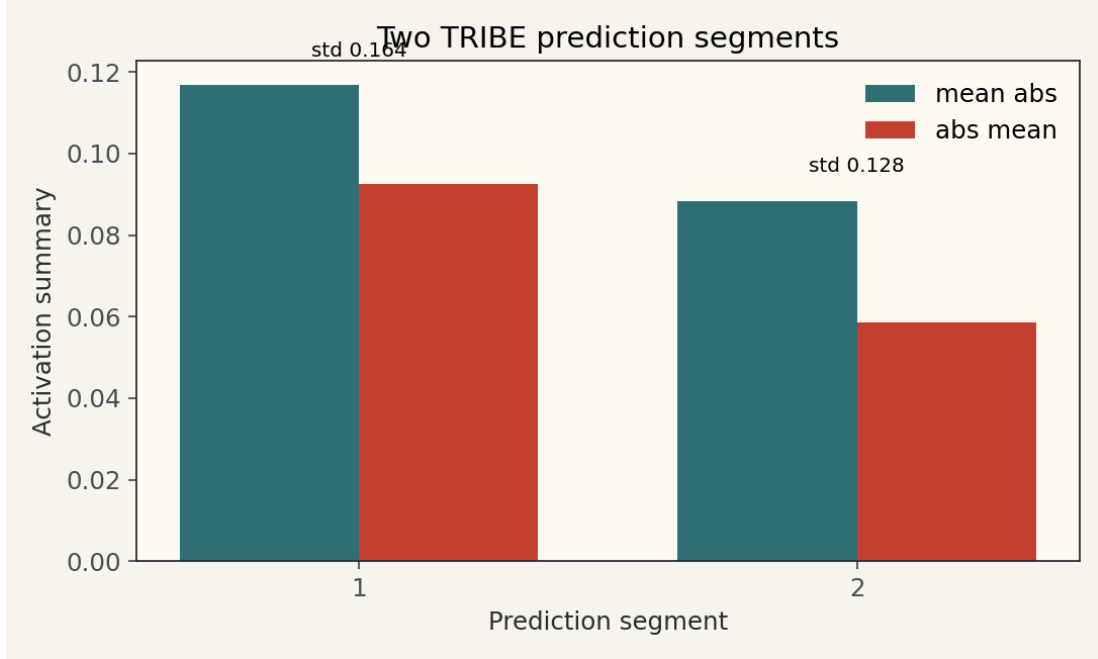


Figure 3: Two retained TRIBES v2 prediction segments. Bars summarize mean absolute predicted response and absolute mean response; labels show segment standard deviation.

structurally valid model-predicted response, and validate the artifact layout. It also provides a concrete visualization path that preserves caveats in the figure caption and metadata.

The response summary is finite and internally serializable. The mean absolute predicted response for the retained stimulus is 0.102670. The two index-half means are 0.102728 and 0.102612; these are serialization summaries, not evidence of biological hemispheric symmetry. With only one completed model prediction, pairwise response similarity is undefined and latent-component analysis is correctly marked as insufficient-samples in the run metrics.

5 Discussion

The main value of this run is methodological. It establishes an end-to-end local path from generated stimulus through TRIBES v2 inference, artifact validation, HTML visualization, and LaTeX manuscript reporting. Because the first-suite run is capped at one prediction, it is cheap enough to repeat on a well-equipped workstation while still exercising the real model backend.

The brain schematic fills an immediate review need: it lets a reader see how a 20,484-length cortical vector is being split and displayed. It does not establish biological plausibility, hemispheric symmetry, or ROI mapping. The correct next step for anatomical reporting is to add the matching fsaverage5 surface coordinates and atlas annotations as declared, versioned inputs, then derive ROI summaries from those files. Until that is done, the diagram should remain labeled as a vertex-index schematic.

TRIBE v2 predicted response on fsaverage5-index schematic

Color encodes mean absolute model-predicted response across two retained output segments.

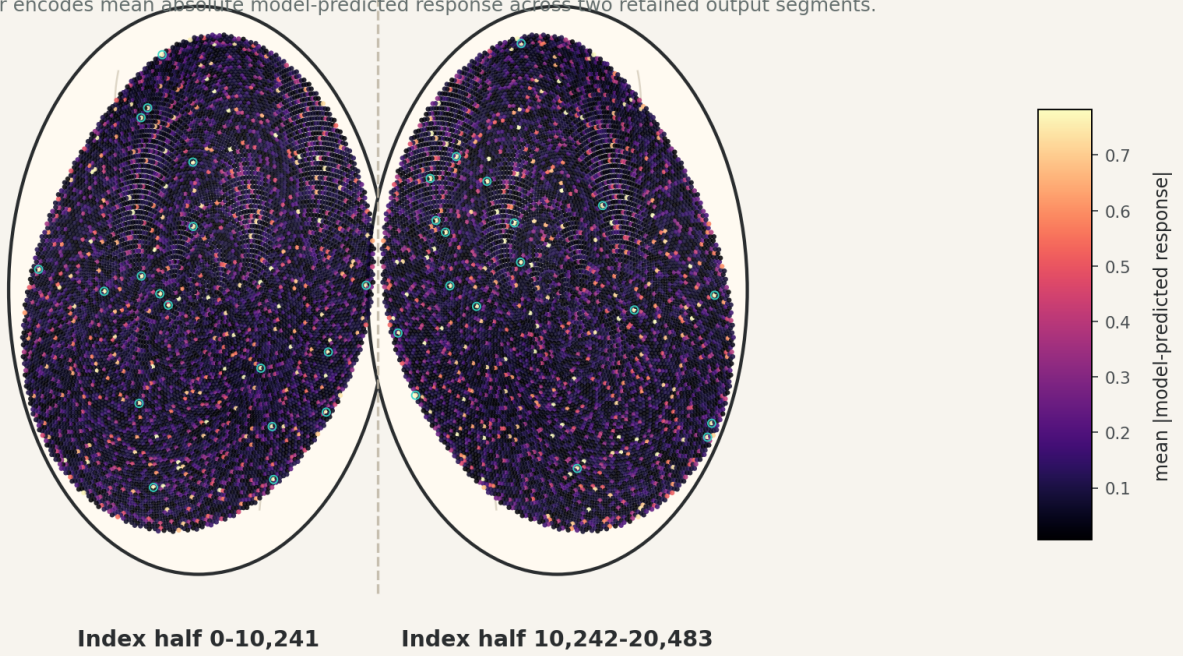


Figure 4: Vertex-index brain schematic. The first half of the 20,484 vector is drawn as one hemisphere-shaped index map and the second half as the other by convention. This is a review diagram for orientation and QA, not a subject-specific anatomical registration, ROI atlas, or biological hemispheric analysis.

6 Limitations

This preprint has strong limitations by design. First, the values are model-predicted average-subject fMRI-like responses, not measured neural activity. Second, the experiment reports one completed prediction, so it cannot support comparisons among stimulus families, statistical inference, or latent network analysis. Third, the brain diagram is not an anatomical registration: no fsaverage5 coordinate file, FreeSurfer subject, surface triangles, or ROI annotation was present in the run artifact. Fourth, the response is based on a generated synthetic visual stimulus and should not be generalized to natural image responses. Fifth, the public TRIBE v2 weights and software have their own non-commercial licensing constraints [3, 2]. Sixth, local inference speed depends on cached model assets and hardware; this run encoded four video feature batches in several minutes on the local machine.

7 Reproducibility

The run was produced from repository commit `ad146aba73d01a6f197f6f2a508c800651406d5f`. The manuscript branch is later because it adds this preprint and prior BOLD5000 preprint infrastructure. The experiment seed is 20260508. Table 3 records the run and model provenance needed to audit the artifact.

The active Python environment used Python 3.12.13 with `braindough==0.1.0`, `tribev2==0.1`

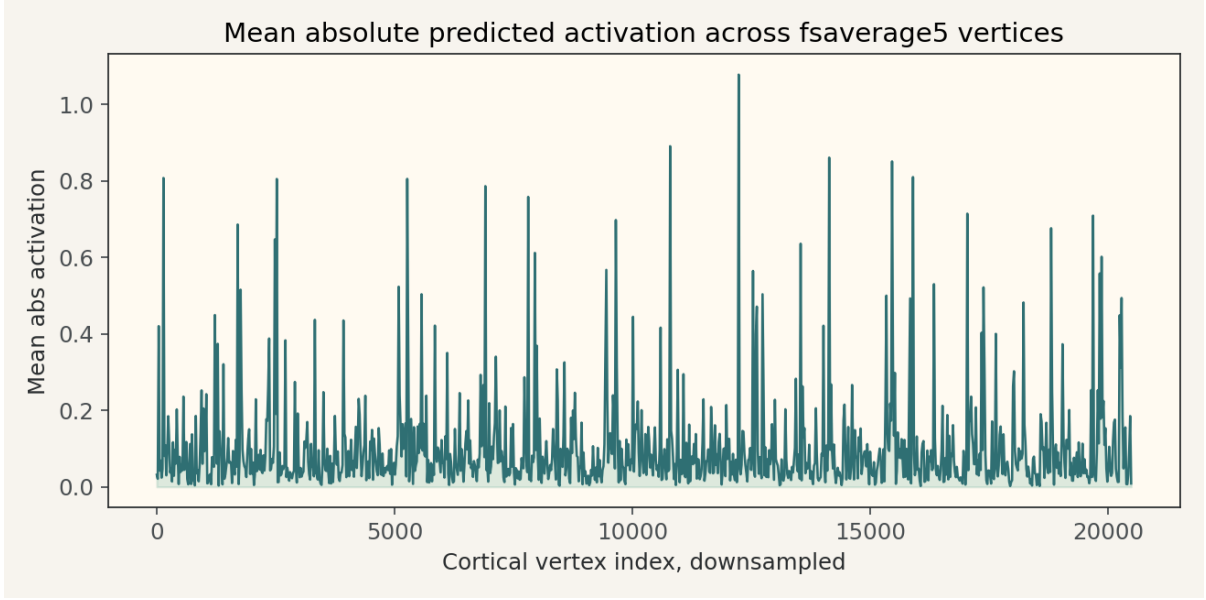


Figure 5: Mean absolute predicted response by downsampled vertex index. This figure is useful for detecting gross serialization or plotting errors, but it is not an anatomical ROI summary.

Table 3: Provenance identifiers for the source run and local model assets.

Field	Value
Run config SHA-256	f4c69bbd3dac5c428b80ee83529ea6c5d0fe985432c5fbe982b75036460ae7de
Response archive SHA-256	5fbbf8041325b1fcac359b52584763d77af19ef0f1ce8af6b2c41417f5054edd
Hugging Face snapshot best.ckpt SHA-256	f894e783020944dcd96e5568550afe2aa9743f9f9c79ffff6b642b7b0c71d558c935fb3fa33f2788bfb509feead94fafbba2f321
config.yaml SHA-256	a17212a257dfd91424e20a5fa8640d107fbbf79e842b6b2d72d17d24399dc9ef

.0, torch==2.6.0, neuralset==0.0.2, numpy==2.2.6, pandas==3.0.2, imageio==2.37.3, imageio-ffmpeg==0.6.0, and pillow==11.3.0.

Structural artifact validation checks the presence and internal consistency of manifest/config files, response metadata, response shapes, required tables, checksums, reports, and finite artifact values. It does not test whether the TRIBE v2 model is accurate, calibrated, or valid for any neuroscientific claim.

The experiment can be rerun from the repository root with:

```
just bootstrap
just storage-doctor
just run-tribe
just artifact-validate RUN_DIR=<new-run-dir>
```

The manuscript PDF is built with:

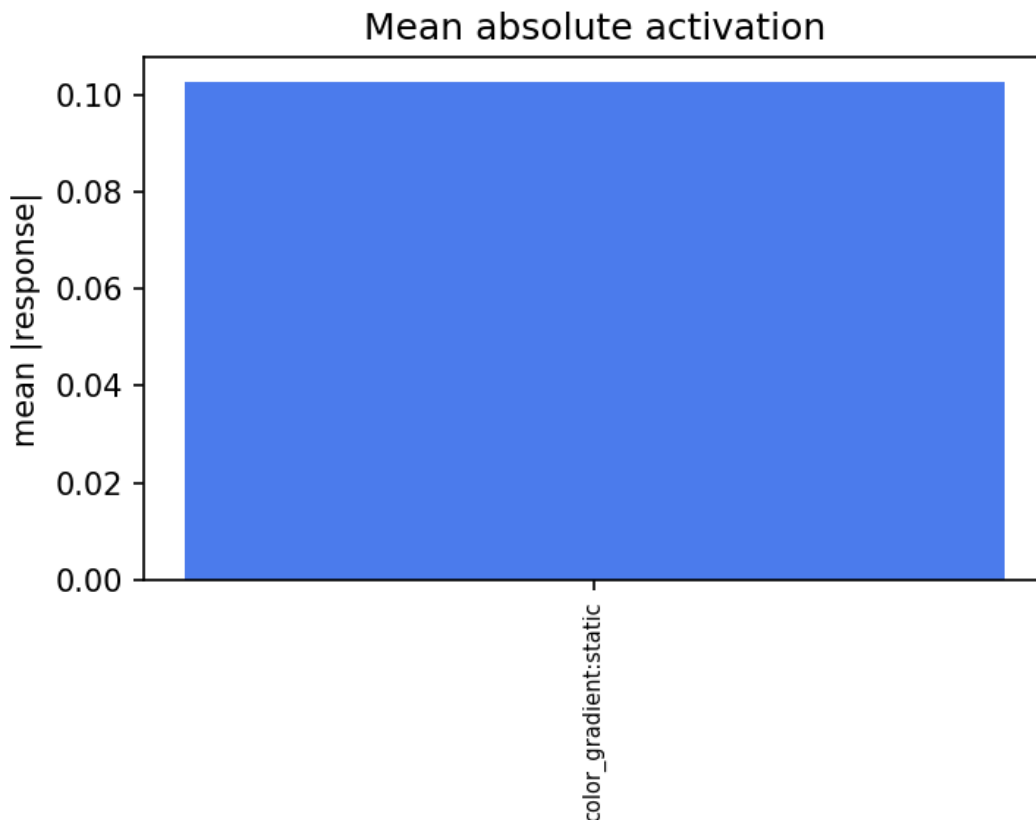


Figure 6: Repository-generated mean absolute predicted-response summary for the one completed TRIBE v2 model prediction.

```
just preprint-build docs/preprints/2026-05-16T233000Z-tribe-v2-schematic-response
```

The committed manuscript directory contains only small derivative figures and tables. The full run directory, model caches, and response archive remain under `BRAINDOUGH_HOME`. This separation keeps the repository reviewable without losing reproducibility metadata.

8 AI-Assisted Review

ChatGPT Extended Pro was used through the browser for iterative critique of scientific framing, overclaim risk, and manuscript structure. Its feedback was treated as advisory. The final claims are restricted to primary sources and the validated local run artifact.

9 Conclusion

A bounded BRAINDOUGH TRIBE v2 run can produce a structurally valid model-predicted response artifact and a reviewable brain-schematic visualization without overclaiming anatomical registration. The result is a practical template for future local in-silico experiments: keep real model runs capped, preserve heavy artifacts outside the repository, commit small derivatives, and label visualizations according to the spatial information actually present.

A Suite Summary

Table 4: Stimuli and completed responses by suite.

Suite	Stimuli	Responses
image activation	3	1
visual controls	5	0
visual perturbations	18	0
temporalization	9	0
audio controls	4	0

B Copied Artifact Files

- tables/metrics.json
- tables/vertex_index_schematic_metadata.json
- tables/response_metrics.csv
- tables/stimuli.csv
- figures/tribe_brain_schematic.png
- figures/tribe_response_distribution.png
- figures/tribe_timepoint_summary.png
- figures/tribe_vertex_profile.png
- figures/mean_abs_activation.png
- figures/response_similarity.png
- figures/color_gradient.png

References

- [1] d’Ascoli, S., Rapin, J., Benchetrit, Y., Brooks, T., Begany, K., Raugel, J., Banville, H., and King, J.-R. A foundation model of vision, audition, and language for in-silico neuroscience. *arXiv preprint* arXiv:2605.04326 (2026). <https://arxiv.org/abs/2605.04326>.
- [2] facebookresearch. TRIBE v2 source repository and README. GitHub (2026). <https://github.com/facebookresearch/tribev2>.
- [3] AI at Meta. facebook/tribev2 model files and license. Hugging Face (2026). <https://huggingface.co/facebook/tribev2>.
- [4] Fischl, B., Sereno, M. I., and Dale, A. M. Cortical surface-based analysis. II: Inflation, flattening, and a surface-based coordinate system. *NeuroImage* 9(2), 195–207 (1999). doi:10.1006/nimg.1998.0396.

[5] FreeSurfer Wiki. FsAverage. <https://freesurfer.net/fswiki/FsAverage>.